

CO5 – AT2

Pseudocode Writing Task (Algorithm Design)

Q2. Subset Sum Problem

SOURCE CODE:

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# Subset Sum Problem using Backtracking

def subset_sum(numbers, target, index, subset, current_sum):

    # Target found
    if current_sum == target:
        print("Subset:", subset)
        return True

    # Stop if sum exceeds target
    if current_sum > target or index == len(numbers):
        return False

    # Include current number
    subset.append(numbers[index])

    if subset_sum(numbers, target, index + 1, subset,
        current_sum + numbers[index]):
        return True

    subset.pop()

    # Exclude current number
    if subset_sum(numbers, target, index + 1, subset,
        current_sum):
        return True

    return False

# Input
numbers = list(map(int, input("Enter the numbers: ").split()))
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target = int(input("Enter target sum: "))
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# Find subset
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if not subset_sum(numbers, target, 0, [], 0):
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    print("No subset found.")
```